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# Mechanistic Validity: A Measurement-Theoretic Framework for Evaluating Interpretability Claims

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## Abstract

Mechanistic interpretability (MI) faces a measurement crisis: 99.7% of sparse autoencoder features are unstable across training runs, attribution patching correlates at  $r = 0.006$  with ground truth on synthetic tasks, two of six standard SAE evaluation metrics fail basic diagnostic tests, and 93% of published MI papers fail execution-grounded reproducibility checks. These findings undermine any claim built on current methodology—including safety-critical claims about deception detection and model monitoring. We present MECHVAL, a measurement-theoretic framework for evaluating MI claims. The framework organizes 27 operational criteria across five validity types (construct, measurement, internal, external, interpretive) in dependency order, assigns claims to one of five verdict tiers (Proposed through Validated), and supports evaluation via 84 metrics spanning five evidence families (causal, structural, behavioral, representational, information-theoretic). The framework is method-agnostic: it applies uniformly to circuits, SAE features, probing classifiers, steering vectors, transcoders, and crosscoders. We describe the framework’s theoretical foundations—drawn from philosophy of science, neuroscience, psychometrics, pharmacology, and causal inference—and its open-source implementation, which includes 54 evaluation tasks, 14 calibration gates, and 12 pre-registered mechanistic claim specifications. Application to 13 published circuits demonstrates that the tier system discriminates: claims range from Proposed (Knowledge Neurons, Probing Classifiers, Gender Bias Circuits) through Causally Suggestive (Grokking, Successor Heads, SAE Features) and Mechanistically Supported (IOI Circuit, Greater-Than, Copy Suppression) to Triangulated (Induction Heads), with verdict differences traceable to specific missing evidence.

## 1 Introduction

When should you trust an interpretability claim? A researcher reports that GPT-2 uses a six-role circuit to solve indirect object identification (?), that sparse autoencoder features correspond to human-interpretable concepts (?), or that a linear probe reveals syntactic structure in transformer activations (?). Each claim comes with experimental evidence—ablation studies, activation visualizations, accuracy numbers. But how well-supported is the claim, really?

Recent work provides uncomfortable answers. ? and ? demonstrate that only 197 of 65,536 sparse autoencoder features—0.3%—are stable across independent training runs. The remaining 99.7% are training artifacts. Any “deception feature” identified using standard SAE methods has a 99.7% prior probability of being an artifact, not a genuine representation.

The field’s primary causal tool fares no better. ? show that attribution patching—the most widely used method for identifying causally relevant components—correlates at  $r = 0.006$  with ground truth on controlled synthetic tasks. This is not a weak correlation; it is no correlation. The method that the field treats as the gold standard for causal attribution provides essentially random rankings of component importance.

The evaluation metrics used to assess these tools are themselves unreliable. The SAEBench audit subjects six standard SAE evaluation metrics to three diagnostic tests: reseed stability, ground-truth correlation,

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and discriminability. Two of six—Token Prediction Probability (TPP) and Sparse Cross-entropy Recovery (SCR)—fail all three diagnostics. TPP exhibits reseed coefficients of variation between 16% and 39%.

Finally, the research itself does not reproduce. MechEvalAgent (Bai et al., 2026) performs execution-grounded evaluation of published MI research: 93% of papers fail reproducibility checks when their code is actually run, and 80% fail coherence checks between claims and evidence. Narrative review—the kind performed by peer reviewers—missed 51 issues that execution-grounded evaluation caught.

The implication is direct: any claim of the form “we found a feature that detects deception” is built on this foundation. If the feature was identified using SAE methods, it is probably a training artifact. If it was validated using attribution patching, the validation is meaningless. If its quality was assessed using TPP or SCR, the assessment is unreliable. If the result was reported in a paper, there is a 93% chance the code does not reproduce.

This paper presents a framework for addressing this crisis. The MECHVAL framework does not propose new detection methods. It provides a systematic protocol for determining whether any detection method—existing or future—actually works. The framework draws on established validation methodology from five fields (philosophy of science, neuroscience, psychometrics, pharmacology, and causal inference) to define 27 operational criteria, organized across five validity types in dependency order, with five verdict tiers that represent qualitative transitions in evidential status.

The framework is method-agnostic. It evaluates SAE features, neural circuits, probing results, steering vectors, transcoders, and crosscoders using the same criteria. It does not privilege one approach over another. It asks the same questions of each: Is the construct defined? Is the evidence causal? Does the finding generalize? Is the measurement reliable? Is the interpretation justified?

We describe the framework (??), its benchmark implementation (??), its application to 13 published circuits (??), and discuss its implications for the field (??).

## References

Bai, X., Baumgartner, T., Sun, C., Holtzman, A., and Tan, C. MechEvalAgent: Execution-grounded evaluation of mechanistic interpretability research. *Preprint*, 2026.